Dual-Branch Dynamic Modulation Network for Hyperspectral and LiDAR Data Classification

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***Abstract*— Deep learning algorithms that can effectively extract features from different modalities have achieved significant performance in multimodal remote sensing (RS) data classification. However, we actually found that the feature representation of one modality is likely to affect other modalities through parameter back-propagation. Even if multimodal models are superior to their uni-modal counterparts, they are likely to be underutilized. To solve the above issue, a dual-branch dynamic modulation network is proposed for hyperspectral (HS) and light detection and ranging (LiDAR) data classification. Firstly, a novel dynamic multimodal gradient optimization (DMGO) strategy is proposed to control the gradient modulation of each feature extraction branch adaptively. Then, a multimodal bi-directional enhancement (MBE) module is developed to integrate features of different modalities, which aims to enhance the complementarity of HS and LiDAR data. Furthermore, a feature distribution consistency (FDC) loss function is designed to quantify similarities between integrated features and dominant features, which can improve the consistency of features across modalities. Experimental evaluations on Houston2013 and Trento datasets demonstrate that our proposed network exceeds several state-of-the-art multimodal classification methods in terms of fusion classification performance.**

***Index Terms*— Feature fusion, hyperspectral image (HSI), imbalanced multimodal learning, land cover classification, multimodal remote sensing (RS).**

I. Introduction

REMOTE sensing (RS) imagery has been widely used for several application scenarios, such as land cover classification, urban development detection, etc, as a non-contact technique [1]. With the development of advanced RS technology, it is now possible to simultaneously collect descriptions of potential land cover objects from a range of sensors, such as hyperspectral (HS) [2], light detection and ranging (LiDAR) [3], synthetic aperture radar (SAR) [4], [5], multispectral (MS) [6], optical detectors [7], etc. This is particularly significant in RS as different sensors can capture various properties of the same geographical area, aiding in the understanding of the entire scene [8]. For instance, the spectral information from HS images (HSIs) is frequently used to identify materials based on their reflectance values [9], [10]. Similarly, LiDAR can provide elevation information about land cover objects, which is helpful to identify objects made of the same materials [11]. Moreover, cloud cover during image acquisition has an impact on optical RS images, which may lead to information loss. SAR can serve as an effective secondary sensor to solve this problem because of its unique imaging technique and ability to penetrate through clouds [12]. The joint use of multiple modalities allows us to identify or classify land cover with greater specificity and accuracy that cannot be achieved using uni-modal data [13]. This has made multimodal feature fusion and interaction as a critical research topic.

Inspired by the success of deep learning models in computer vision [14], [15], [16], [17], [18], [19], numerous multimodal feature fusion networks have been proposed for RS land cover classification [20], [21], [22], [23], [24]. Ge et al. [21] established a feature fusion framework based on residual connection for HS-LiDAR land cover classification. Xu et al. [22] recommended the use of a standard dual-branch CNN for HS-LiDAR fusion, where one branch extracts spatial and spectral representations from HS data while the other branch extracts LiDAR image features from the cascade network. Wang et al. [23] presented another two-branch classifier that uses 3-D CNN and 2-D CNN to extract HS and LiDAR features, respectively, to exploit the rich spectral information of HSIs. Wang et al. [24] developed a triple-branch classification network to fuse HS and LiDAR images collaboratively. In deep learning methods, a multibranch architecture is typically followed, where each branch corresponds to a type of feature extraction block, and then the features are passed through some fusion strategies (e.g., concatenation [25], alignment [26]) to form a joint representation for further classification.

Feature interaction is another commonly used approach to enhance the representation of multimodal discriminative features. Several studies have demonstrated that it can strengthen the richness of subsequent representations and improve the discriminative ability of the classifier. Weight-sharing strategies were applied in [27] and [28] for feature interaction. Mohla et al. [29] developed spatial attention via LiDAR features to weight HSI features. Liao et al. [30] proposed a method for fusing morphological profiles of HS and LiDAR data on manifolds, which involves utilizing graph-based subspace embedding techniques [31], [32] on a non-Euclidean space. Zhao et al. [33] used octave convolution [34] to subtly distinguish low- and high-frequency information while interacting with features of the same frequency from both branches.

Although these multibranch deep learning architectures have undoubtedly demonstrated impressive performance, they come with significant limitations. Specifically, due to the

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back-propagation of parameters, the feature representation of one modality may influence the feature representation of other modalities. This potential drawback should not be ignored when designing these types of architectures for multimodal fusion tasks. Recent research has shown that multimodal models using joint training strategies to optimize a uniform learning objective for all modalities can sometimes be inferior to uni-modal models [35], revealing an imbalanced multimodal learning problem [35], [36], [37], [38].

To address this problem, some approaches have employed additional classifiers or pretrained models to help the training of multimodal models [39], [40], [41]. Li et al. [39] first trained the feature extraction networks of HS and LiDAR images, respectively, and then included these two pretrained networks in the training of the whole network as a part of the network. However, this step-wise approach requires additional work to train multiple models. Parida et al. [40] proposed a jointly learned modality attention network to predict the dominant modality. Fan et al. [41] used prototypes to enhance clustering and introducing an entropy regularization term to prevent premature convergence of dominant modality.

In this article, we first analyze the imbalanced multimodal learning problem in a dual-branch RS data classification network from an optimization perspective. In order to alleviate this issue, we propose a novel dynamic multimodal gradient optimization (DMGO) strategy to control the gradient modulation of each feature extraction branch. Then, inspired by feature interaction, we design a multimodal bi-directional enhancement (MBE) module to fully account for complementary information between multimodal data. In addition, a new regularization loss function for network training is created to encourage the proposed model to learn shared and consistent information from multimodal features throughout the learning process. Through the MBE module and regularization loss, we hope to propose a multimodal feature extraction framework that can simultaneously take into account multimodal feature complementarity and intraclass feature consistency.

The main contributions of this article are summarized below.

1) In order to solve the imbalanced convergence of dual-branch feature extraction networks, we propose a novel DMGO strategy. It can independently adjust the optimization of each branch during backpropagation and dynamically control gradient modulation, which contributes to the fusion classification of HS and LiDAR data.
2) The MBE module is developed as a plug-and-play module for dual-branch feature representation, which can effectively enhance and integrate information from multiple modalities and improve the performance of the model.
3) A feature distribution consistency (FDC) loss function is proposed to measure the similarity between integrated features and dominant modality features, which makes the representation of each category as consistent as possible.

The rest of this article is structured as follows. Section II describes the proposed framework and algorithm for HS and LiDAR data classification. In Section III, the experimental setup and classification performance are analyzed. Conclusions are finally summarized in Section IV.

II. Methodology

A. Overview

In multimodal RS data classification networks, the modality with superior performance usually dominates the optimization process of multimodal classifiers, resulting in the feature representation of one modality suppressing the features of other modalities. In order to cope with this imbalanced optimization, a dual-branch dynamic modulation network is proposed for HS and LiDAR data classification. The overall architecture of the proposed model is graphically shown in Fig. 1.

First, a dual-branch CNN is developed for feature extraction from HS and LiDAR images as shown in Fig. 2. A novel spatial-spectral residual network composed of the spectral learning section (SPE) and spatial learning section (SPA) is designed to improve the feature extraction performance of HSIs. The precise structure is depicted in Fig. 3. Specifically, the SPE module is designed to extract deep spectral–spatial features from the original HS data using a 3-D-CNN structure. In contrast, the SPA module focuses on extracting spatial features from the same data. To enhance the compatibility of these two modules, a 3-D-CNN layer is added as a connection layer, as shown in Fig. 2(a). By integrating spatial and spectral information, the proposed architecture can extract more discriminative features and achieve better classification performance. Moreover, the SPA and SPE employ the residual structure to enable the network to learn more robust spectral-spatial representations. These modules are crucial components of our proposed end-to-end feature extraction network.

Second, the DMGO strategy is proposed to independently adjust the optimization of each branch during backpropagation and dynamically control gradient modulation. Precisely, the DMGO strategy measures the contribution of each uni-modal component by predicting the classification accuracy of each uni-modal branch. Then the dominant and auxiliary modalities of the current epoch are determined by the contribution ratio (as described in Section II-B). We achieve balanced convergence of the multimodal model by gradient optimization. Third, the MBE module is developed to generate the integrated feature of different modalities. This module enhances the feature extraction of HS and LiDAR data by generating interacted features using self-attention masks from HS features to focus on their spectral-specific traits and cross-attention masks from LiDAR to enhance spatial representation. In addition, the similarity between dominant features and integrated features is measured by a novel FDC loss function. By this regularization constraint, the proposed model is encouraged to learn some consistent information shared by different sources, which can improve the consistency of features across modalities. In the end, pixel-wise classification result is obtained by feeding the final multimodal features into the classification block.

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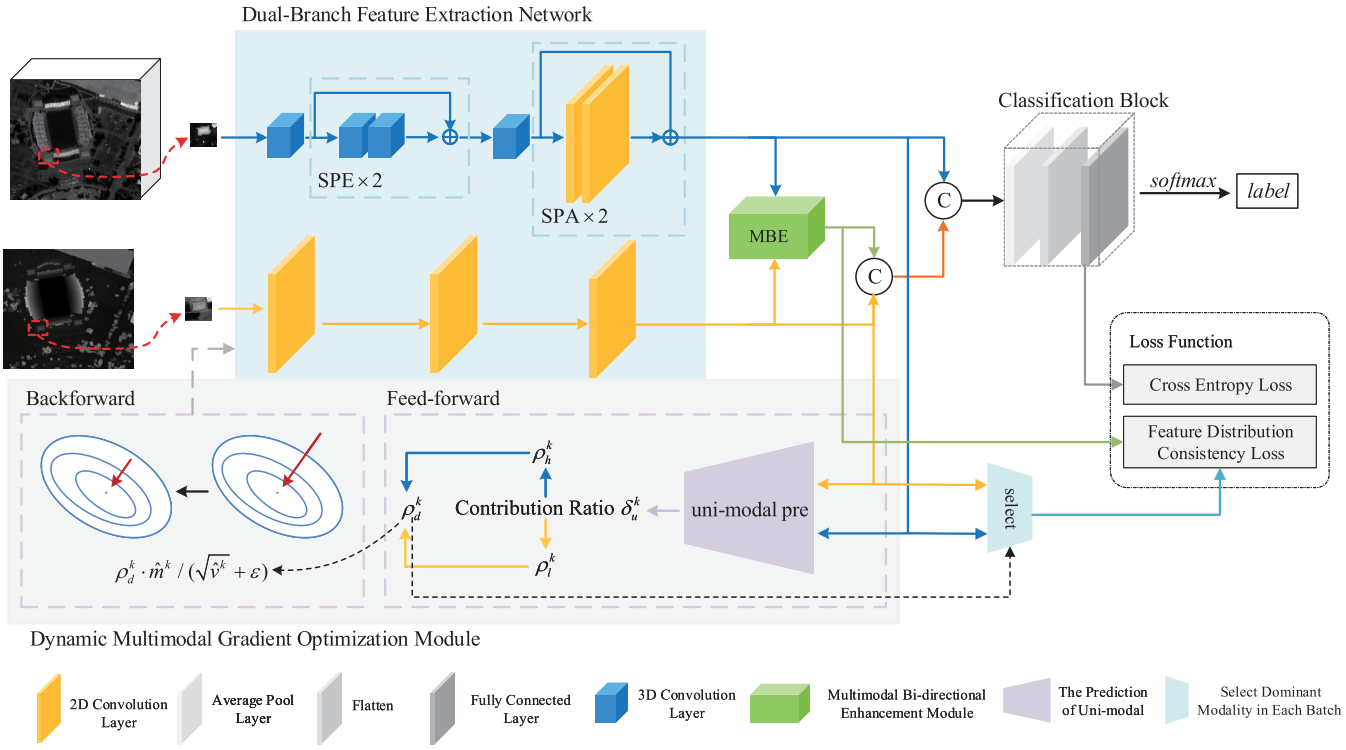


Fig. 1. Overall architecture of the proposed dual-branch dynamic modulation network.

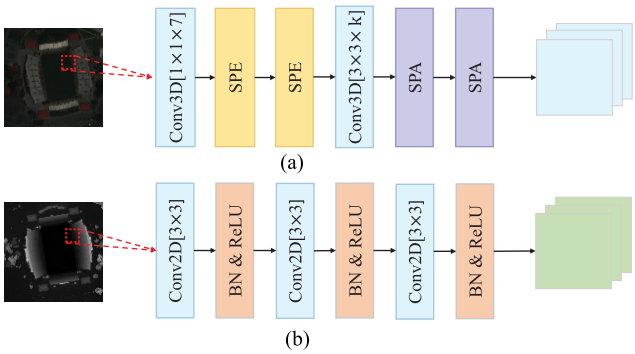


Fig. 2. Dual-branch feature extraction network for HS and LiDAR images. Network structure for (a) HS and (b) LiDAR images.

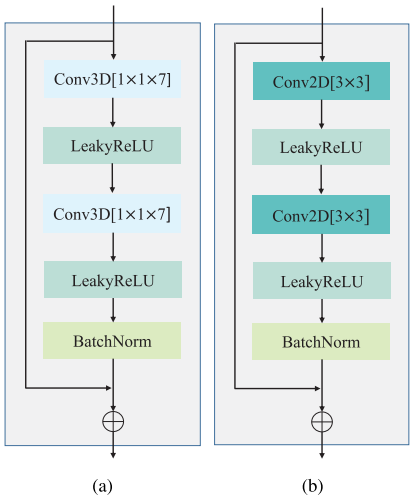


Fig. 3. Network structure of (a) SPE and (b) SPA. SPE, which is designed to extract deep spectral–spatial features from the original HS data using a 3-D-CNN structure. SPA, which focuses on extracting spatial features from the same data. To enhance the compatibility of these two modules, a 3-D-CNN layer is added between them as a connection layer.

B. Dynamic Multimodal Gradient Optimization

For the classification task of multimodal RS images, there is usually a natural imbalance between different modal data. Specifically, each uni-modal often has different properties and tends to converge at different rates during the training process. The feature representation of one modality is likely to affect other modalities through parameter back-propagation. The modality that performs better leads to a more minor joint discriminative loss, thus masters the training process and results in an under-optimized result by propagating a limited gradient over the inferior modality. We analyze the above problem from an optimization perspective and then propose the DMGO strategy to adaptively control gradient modulation during training.

The features of HS and LiDAR patches are extracted using the encoders $\Phi_h(\theta_h, \cdot)$ and $\Phi_l(\theta_l, \cdot)$, respectively, where θ_h and θ_l represent the parameters of the encoder. Then, in order to combine the representations from HS and LiDAR, we used concatenation, which is the most commonly used fusion technique. The variables that affect each modal optimization are analyzed below. Let $W \in \mathbb{R}^{N\times(d_{\Phi_h}+d_{\Phi_l})}$ and $b \in \mathbb{R}^N$ represent the parameters of a linear classifier. Then the logits output of the feature extraction module can be written as

$$f(x_i) = W\left[\Phi_h\left(\theta_h, x_h^i\right); \Phi_l\left(\theta_l, x_l^i\right)\right] + b \tag{1}$$

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where x_h^i and x_l^i are the ith input feature of HS and LiDAR patches, respectively. In order to observe how each modality's features are optimized, W can be represented as the union of the following two blocks: $[W_h, W_l]$. The logits output of the feature extraction module, as shown in (1), can be rewritten as follows:

$$f\left(x^i\right) = W_h \cdot \Phi_h\left(\theta_h, x_h^i\right) + W_l \cdot \Phi_l\left(\theta_l, x_l^i\right) + b. \tag{2}$$

The gradient descent (GD) optimization method is used during training, and the encoder parameters W_h^{k+1} and θ_h^{k+1} are updated as

$$\begin{aligned} W_h^{k+1} &= W_h^k - \eta \nabla_{W_h} L\left(W_h^k\right) \\ &= W_h^k - \eta \frac{1}{M} \sum_{i=1}^{M} \frac{\partial L}{\partial f\left(x^i\right)} \Phi_h\left(\theta_h, x_h^i\right) \end{aligned} \tag{3}$$

$$\begin{aligned} \theta_h^{k+1} &= \theta_h^k - \eta \nabla_{\theta_h} L\left(\theta_h^k\right) \\ &= \theta_h^k - \eta \frac{1}{M} \sum_{i=1}^{M} \frac{\partial L}{\partial f\left(x^i\right)} \frac{\partial\left(W_h^k \cdot \Phi_h^k\left(\theta_h, x_h^i\right)\right)}{\partial \theta_h^k} \end{aligned} \tag{4}$$

where L is the cross-entropy (CE) loss function, η is the learning rate

$$L = \frac{1}{M} \sum_{i=1}^{M} \left(\log \sum_{j=1}^{N} e^{f\left(x^i\right)_j} - \log f\left(x^i\right)_{y^i} \right) \tag{5}$$

where $f(x^i)_j$ is the predicted output for class j.

According to (3) and (4), we can see that the optimization of one modality has little correlation with the other, apart from the term related to the CE loss function $[\partial L/(\partial f(x^i))]$. So the uni-modal feature extraction branches in multimodal models have little ability to modify each other based on back-propagation. Equation (2) states that $[\partial L/(\partial f(x^i))]$ can be expressed as follows:

$$\frac{\partial L}{\partial f\left(x^i\right)_m} = \left(\frac{e^{\left(W_h \cdot \Phi_h^i + W_l \cdot \Phi_l^i + b\right)_m}}{\sum_{j=1}^{N} e^{\left(W_h \cdot \Phi_h^i + W_l \cdot \Phi_l^i + b\right)_j}} - 1 \right)_{m = y^j} \tag{6}$$

where the notations Φ_h^i and Φ_l^i are obtained by simplifying the expressions $\Phi_h^k(\theta_h, x_h^i)$ and $\Phi_l^k(\theta_l, x_l^i)$, respectively. Additionally, y^i denotes the ground truth of the ith input, where $y^i \in \{1, \ldots, C\}$ and C represents the total number of categories.

According to (6), we can infer that the performance for one modality, such as HS, is better, the sample x^i belonging to class y^i contributes more to the gradient $[\partial L/(\partial f(x^i)_{y^i})]$ through $W_h \cdot \Phi_h^i$. Therefore, LiDAR can only obtain limited optimization because of the relatively low confidence for the correct class. This occurrence shows that the performance-enhancing modality controls the progress of optimization, and we name this modality the dominant modality. In contrast, we name another modality as the auxiliary modality. Thus, as the multimodal model training approaches converge, the auxiliary modality may still have the problem of under-optimized representation and requires further training.

In order to cope with this imbalanced optimization, we modulate the parameter gradients during the training of each feature extraction branch by DMGO strategy.

We design s_h^i and s_l^i serve as the approximate prediction of uni-HS and uni-LiDAR for a multimodal model, respectively,

$$s_h^i = \sum_{n=1}^{N} 1_{n=y_i} \cdot \operatorname{softmax}\left(W_h^k \cdot \varphi_h^k\left(\theta_h, x_h^i\right) + \frac{b}{2} \right)_n \tag{7}$$

$$s_l^i = \sum_{n=1}^{N} 1_{n=y_i} \cdot \operatorname{softmax}\left(W_l^k \cdot \varphi_l^k\left(\theta_l, x_l^i\right) + \frac{b}{2} \right)_n. \tag{8}$$

We design the contribution ratio δ_h^k as

$$\delta_h^k = \frac{\sum_{i \in B_k} s_h^i}{\sum_{i \in B_k} s_l^i} \tag{9}$$

where δ_l^k is accordingly defined as the reciprocal of δ_h^k. If $\delta_h^k > 1$, then the corresponding modality, HS, is defined as the dominant mode at the current moment, and LiDAR is defined as the auxiliary modality (Similarly when $\delta_l^k > 1$). By dynamically monitoring the contribution differences between modalities and adapting, we achieve adaptive optimization of gradient via

$$\rho_u^k = \begin{cases} 1 - \tanh\left(\alpha \cdot \delta_u^k\right), & \delta_u^k > 1 \\ 1, & \text{others} \end{cases} \tag{10}$$

where α is a hyper-parameter to regulate the level of optimization. By integrating the coefficient ρ_u^k into the adaptive moment estimation (Adam) [42] optimization method, the update formula of θ_u^k in iteration k is as follows:

$$\theta_u^{k+1} = \theta_u^k - \eta \cdot \rho_u^k \cdot \hat{m}^k \Big/ \left(\sqrt{\hat{v}^k} + \varepsilon \right) \tag{11}$$

where $\hat{m}^k$ denotes the first moment estimation of gradient, $\hat{v}^k$ is the second moment estimation of gradient, ε is a constant added to maintain numerical stability.

The dominant modality and auxiliary modality are selected dynamically during training. Utilizing ρ_u^k, the optimization of the dominant modality is limited, while the auxiliary modality is not affected and can minimize inhibited optimization.

C. Multimodal Bi-Directional Enhancement Module

In multibranch feature extraction networks, features from each uni-modal are highly complementary due to the heterogeneity of HS and LiDAR data. Inspired by feature interaction, we design the MBE module to enhance and integrate information from different modalities. As shown in Fig. 4, the MBE module enhances the feature extraction of HS and LiDAR data by generating interacted features using self-attention masks from HS features to focus on their spectral-specific traits and cross-attention masks from LiDAR to enhance spatial representation. The calculation process of the MBE module for the multimodal integrated feature is defined as follows.

A local feature map of HS data $F_h \in \mathbb{R}^{C \times H \times W}$, where C is the number of channels and $H \times W$ is the number of pixels, is fed into three convolution layers, g_1, θ_1, ϕ, respectively, to generate three new feature maps $\{F_{h,a}, F_{h,b}, F_{h,c}\}$, where $\{F_{h,a}, F_{h,b}\} \in \mathbb{R}^{C/8 \times H \times W}$, $\{F_{h,c}\} \in \mathbb{R}^{C \times H \times W}$. The first one is then reshaped to become $(H \times W) \times C/8$, the second one to become $C/8 \times (H \times W)$, and the final one to become $C \times (H \times W)$. We calculate the spatial attention matrix of

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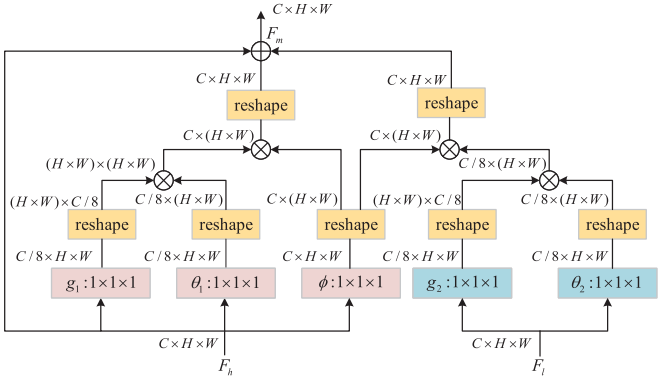


Fig. 4. Illustration of the MBE module. F_h and F_l denote the features extracted from HS and LiDAR patches, respectively, while F_m represents the integrated multimodal feature obtained via the MBE module. The module uses self-attention masks from HS features to focus on their own spectral-specific traits and cross-attention masks from LiDAR to enhance spatial representation. The integrated features are then combined to produce the final multimodal feature representation.

HS data S_h^{ji} by a softmax layer and a matrix multiplication between $F_{h,a}$ and $F_{h,b}$

$$S_h^{ji} = \frac{\exp\left(F_{h,a}^i \cdot F_{h,b}^j\right)}{\sum_{i=1}^{P} \exp\left(F_{h,a}^i \cdot F_{h,b}^j\right)} \tag{12}$$

where $P = H \times W$, S_h^{ji} measures the ith position's impact on jth position.

Similarly, we generate three more feature maps $\{F_{l,a}, F_{l,b}, F_{l,c}\}$ from a local feature map of LiDAR data $F_l \in \mathbb{R}^{C\times H\times W}$. Then $F_{l,a}$ and $F_{l,b}$ are reshaped to $(H \times W) \times C/8$ and $C/8 \times (H \times W)$, respectively. The spatial attention matrix of LiDAR data S_l^{ji} is calculated using a softmax layer and a matrix multiplication between $F_{l,a}$ and $F_{l,b}$

$$S_l^{ji} = \frac{\exp\left(F_{l,a}^i \cdot F_{l,b}^j\right)}{\sum_{i=1}^{P} \exp\left(F_{l,a}^i \cdot F_{l,b}^j\right)}. \tag{13}$$

Finally, we obtain the multimodal integrated feature as follows:

$$F_m^j = \lambda_1 \sum_{i=1}^{P} \exp\left(F_{h,c}^i \cdot S_h^{ji}\right) + \lambda_2 \sum_{i=1}^{P} \exp\left(F_{h,c}^i \cdot S_l^{ji}\right) + F_h^i \tag{14}$$

where λ_1 and λ_2 are taught to gradually assign heavier weights after being initialized at zero [43].

D. FDC Loss Function

As feature vectors extracted from several modalities lie in different representation spaces, interactions or correlations may not only exist among the corresponding dimensions of multimodal vectors. Thus, only use concatation as the fusion strategy may affect fusion performance. To enhance the efficiency of the proposed model, we proposed a novel regularization loss function to encourage the model to learn shared and consistent information from multimodal features throughout the learning process. We want the features, especially some more essential features, i.e., dominant features and integrated features, to have a similar spatial distribution.

The FDC loss function for network training is proposed, which aims to constrain the distribution of the multimodal integrated features to be similar to the distribution of the dominant features. Specifically, we limit the distance between two feature maps by Kullback–Leibler (KL) divergence. In other words, if the dominant features of some patches are packed in the feature space, the corresponding integrated objects are expected to be close in the feature space. The following equation can be used to calculate the FDC loss function:

$$L_{\text{FDC}} = \text{KL}(\text{avg}(M \cdot F_m), F_d) \tag{15}$$

where M denotes the prediction of a local image, F_m denotes the multimodal integrated feature, F_d denotes the dominant modal feature, avg denotes the average pooling operation.

The total loss function to optimize the proposed dual-branch dynamic modulation model can be written as follows:

$$L = L_{\text{CE}} + \gamma L_{\text{FDC}} \tag{16}$$

where γ is hyper-parameter.

The FDC loss can boost the ability of our proposed model to capture more subtle differences between classes and achieve better classification results.

Algorithm 1 Train the Proposed Dual-Branch Dynamic Modulation Network

Input: Remote sensing images and labels $\text{P} = \{(x_h^i, x_l^i), y^i\}_{i=1,2,\ldots M}$, hyper-parameters α, η, γ, iteration number K

Output: The trained proposed model.

for $k = 0, 1, \ldots, K-1$ **do**
- Sample a fresh batch B_k from P ;
- Obtain the features of HS data F_h and LiDAR data F_l respectively by feeding forward B_k to the Feature Extraction Module ;
- Calculate the contribution ratio δ_h^k by Equation (7), (8) and (9) ;
- Calculate ρ_u^k and determine the dominant modal feature F_d by Equation (10) ;
- Calculate $\hat{m}^k$ and $\hat{v}^k$ by back-propagation ;
- Update back propagation gradient by Equation (11) ;
- Obtain the multimodal integrated feature F_m by MBE module ;
- Construct a novel feature distribution consistency loss L_{FDC} by Equation (15) ;
- Optimize the proposed model by Equation (16) ;

end

III. Experiments

A. Datasets

1) Houston2013 Data: The dataset was introduced in the 2013 IEEE GRSS Data Fusion contest. The images document the University of Houston campus in TX, USA, as well

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TABLE I

HOUSTON2013 DATASET DESCRIPTION, INCLUDING THE CATEGORY NAME AND THE SIZE OF TRAINING AND TESTING SAMPLES. TR SET STANDS FOR TRAINING SET. TS SET STANDS FOR TESTING SET

No	Category Name	TR Set	TS Set	Color
0	Healthy grass	198	1053	
1	Stressed grass	190	1064	
2	Synthetic grass	192	505	
3	Tree	188	1056	
4	Soil	186	1056	
5	Water	182	143	
6	Residential	196	1072	
7	Commercial	191	1053	
8	Road	193	1059	
9	Highway	191	1036	
10	Railway	181	1054	
11	Parking lot 1	192	1041	
12	Parking lot 2	184	285	
13	Tennis court	181	247	
14	Running track	187	473	
-	Total	2832	12,197	-

TABLE II

TRENTO DATASET DESCRIPTION, INCLUDING THE CATEGORY NAME AND THE SIZE OF TRAINING AND TESTING SAMPLES. TR SET STANDS FOR TRAINING SET. TS SET STANDS FOR TESTING SET

No	Category Name	TR Set	TS Set	Color
0	Apple trees	129	3905	
1	Buildings	125	2778	
2	Ground	105	374	
3	Wood	154	8969	
4	Vineyard	184	10317	
5	Roads	122	3252	
-	Total	819	29,595	-

as the rural land of the area. The datasets are composed of two data sources and both of them consist of 1905 × 349 pixels. The HSI is available in 144 bands with wavelengths ranging from 364 nm to 1046 nm and a spectral interval of 10 nm. For the same image area, height information is provided by the LiDAR data in only one band. Moreover, field surveys were conducted for 15 LULC-related classes. The category names are shown in Table I, along with specific information on the size of the training and test samples. Fig. 5(a) shows the ground truth map used for training and testing classification, as well as a false-color photograph of the scene under study.

2) Trento Data: The rural area of South Trento is where the relevant scenes were taken. Two data sources make up these datasets. The Optech ALTM 3100EA imaging sensor only has one band, which was used to acquire the LiDAR image. The HSI was obtained by the AISA Eagle imaging sensor and consisted of 63 bands. Both of them have a spatial resolution of 1 m and 600 × 166 pixels. The complete scenario is divided into six categories, which are described in Table II and represented visually in Fig. 5(b).

B. Experimental Setup

1) Assessment Criteria: In our experiment, three commonly used indices, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient, are computed to assess the efficacy of multimodal RS land cover classification. An improved classification execution is indicated by a more excellent OA value and Kappa coefficient when the classification results are compared to the ground truth maps.

TABLE III

CONFIGURATION OF THE NETWORK USED FOR EXTRACTING FEATURES FROM HS DATA, WHERE C REPRESENTS THE NUMBER OF CATEGORIES IN THE DATASET

Layer Name	Kernel Size	Output Size for Houston2013	Output Size for Trento
Input		11×11×144	7×7×63
Conv 1	1×1×7	11×11×46×28	7×7×19×28
SPE	1×1×7	11×11×46×28	7×7×19×28
Conv 2	3×3×k	k=46, 9×9×28	k=19, 5×5×28
SPA	3×3	9×9×28	5×5×28
MBE		9×9×28	5×5×28
Fusion		1×1×C	1×1×C

2) Comparative Analysis With State-of-the-Art Multimodal Fusion Algorithms: To demonstrate the viability of the proposed model, several conventional and cutting-edge multimodal fusion networks were chosen for quantitative and qualitative comparison; including typical two-branch model (e.g., Two-Branch CNN [22]), concatenation-based model (e.g., MDL-Late [25], MDL-Middle [25]), encoder-decoder model (e.g., EndNet [44], interleaving perception convolutional neural network (IP-CNN) [45]), attention based model (e.g., S^2ENet [46]), multiattentive hierarchical fusion net (MAHiDFNet) [24] and CNN-PPF model [47].

3) Implementation Details: The proposed dual-branch dynamic modulation model is implemented on the Pytorch platform. The original multimodal data are input into the proposed model and trained end-to-end. Adam acts as an optimizer and modifies the gradient of the model throughout the training phase. The input patch size of the Trento dataset in the model is set to 7 × 7 pixels, and the batch size is 64. The input patch size of the Houston2013 dataset in the model is set to 11 × 11 pixels, and the batch size is 64. A cosine annealing approach is used to modify the learning rate, which is initially set to 0.001. All algorithms are trained uniformly with 128 epochs. In the DMGO strategy, the hyper-parameter α is selected from the range [0, 1] according to the validation set. In the MBE module, the hyper-parameter γ is set to 0.1 on Houston2013 data and 0.3 on Trento data. Meanwhile, in the process of data preprocessing, all feasible multiple 90° rotations and flips are used to expand the training samples, thus improving the robustness of the proposed model. The specific network configuration for Houston2013 and LiDAR datasets is shown in Tables III and IV.

4) Parameter Tuning: The design of the network structure affects the effectiveness of the multimodal feature fusion model. This section describes how the proposed model performs when the network's hyper-parameters are altered. These hyper-parameters include the input patch size, weight parameter γ of the loss function, weight parameter α of gradient modulation and network learning rate.

The information content of each input is determined by the input patch size, and the performance of feature extraction is affected to some extent by the quantity of information.

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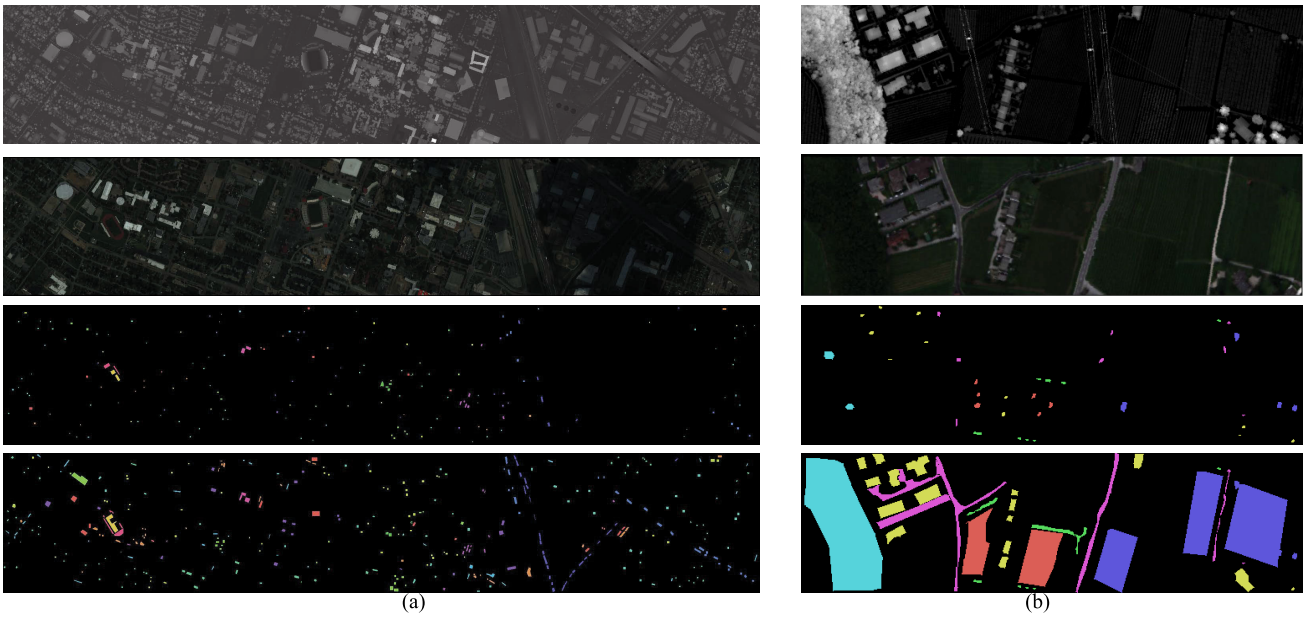


Fig. 5. Multimodal RS datasets with different labeled classes. (Top to bottom) Gray-scale images of LiDAR data, false color images of HS data (Houston2013 dataset with bands 59, 40, and 23, Trento dataset with bands 25, 15, and 5), ground truth maps for training and ground truth maps for testing. (a) Houston2013 dataset. (b) Trento dataset.

TABLE IV

CONFIGURATION OF THE NETWORK USED FOR EXTRACTING FEATURES FROM LiDAR DATA, WHERE C REPRESENTS THE NUMBER OF CATEGORIES IN THE DATASET

Layer Name	Kernel Size	Output Size for Houston2013	Output Size for Trento
Input		11×11×1	7×7×1
Conv 1	3×3	11×11×8	7×7×8
Conv 2	3×3	11×11×16	7×7×16
Conv 3	3×3	9×9×28	5×5×28
MBE		9×9×28	5×5×28
Fusion		1×1×C	1×1×C

We experimentally give the classification performance of the models with different input patch sizes on two real-world datasets to determine the effect of the input patch size on the model. Fig. 6(a) displays the land cover classification maps for two real-world RS datasets with varying patch sizes (ranging from 3 × 3 to 15 × 15).

An appropriate learning rate plays a critical role in deep learning network training as it governs the rate of convergence toward the objective function's minimum. This ensures that the loss function reaches its minimum within a finite number of iterations. Fig. 6(b) illustrates the proposed network's classification performance in the two datasets using various learning rate variations.

In multibranch feature extraction networks, features from each uni-modal are extracted independently through the corresponding branch. We propose the FDC loss function to constrain the distribution of the multimodal integrated feature to be similar to the dominant features in Section II-D. The weight parameter γ is used to measure the balance between the FDC loss and the CE loss. The weight parameter α aim to control the degree of gradient modulation. Fig. 6(c) and (d)

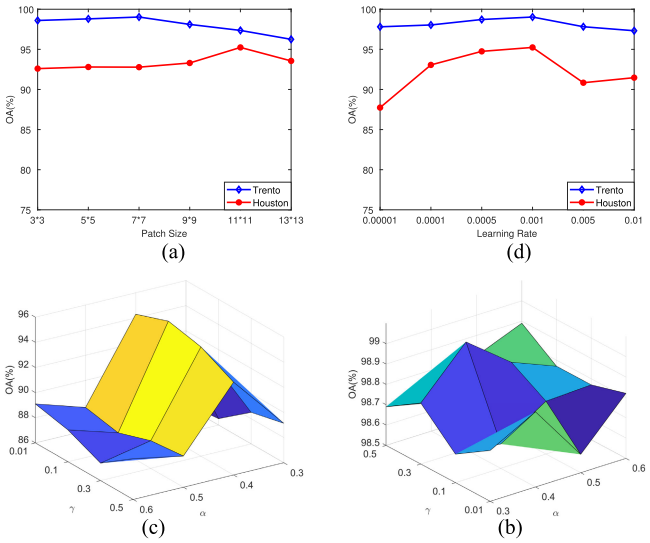


Fig. 6. OA (%) obtained by our dual-branch dynamic modulation network with different hyper-parameters on Houston2013 and Trento datasets. Comparisons of model performance with different (a) input patch sizes and (b) learning rates. Performance comparisons on (c) Houston2013 dataset and (d) Trento dataset by varying the hyper-parameters α and γ.

shows the effect of OA on land cover classification results by changing weight parameters γ and α on the two datasets.

C. Classification Results and Analyses

To assess the effectiveness of our dual-branch dynamic modulation model, we compare it with several advanced multimodal classifiers, including Two-branch CNN [22], MDL-Late [25], MDL-Middle [25], EndNet [44], CNN-PPF [47], IP-CNN [45], S^2ENet [46] and MAHiDFNet [24]. The classification outcomes of several multimodal algorithms on Houston2013 dataset and Trento dataset are quantitatively listed in Tables V and VI, respectively.

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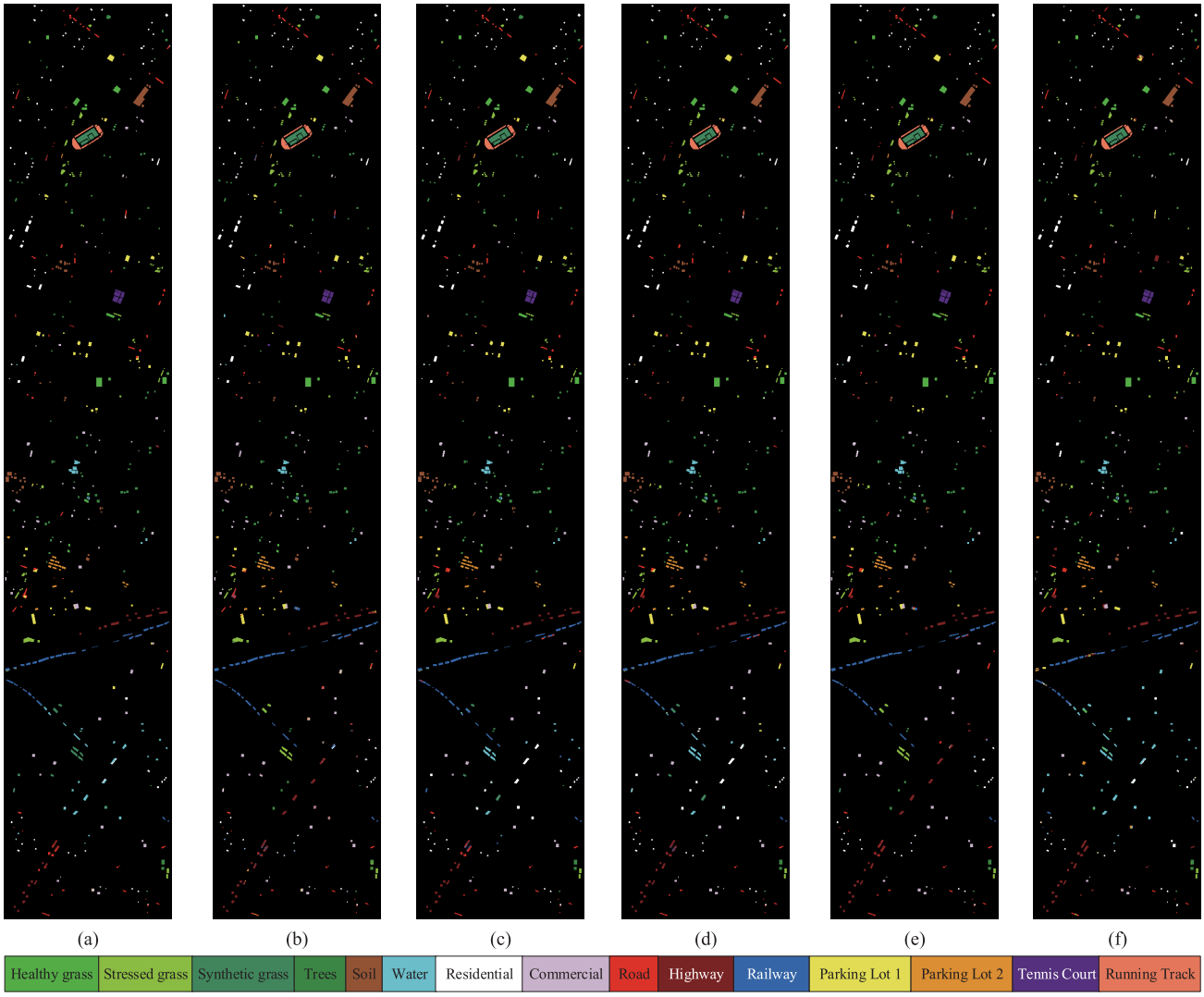


Fig. 7. Classification maps for Houston2013 dataset. (a) Two-branch CNN. (b) EndNet. (c) MDL-Middle. (d) MDL-Late. (e) S^2ENet. (f) Proposed.

TABLE V
QUANTITATIVE COMPARISON ON HOUSTON2013 DATASET OF DIFFERENT METHODS. BEST OUTCOMES ARE DENOTED IN BOLD

Category	Two-Branch CNN [22]	EndNet [44]	CNN-PPF [47]	MDL-Middle [25]	MDL-Late [25]	IP-CNN [45]	S^2ENet [46]	MAHiDFNet [24]	Proposed
Healthy grass(C0)	83.1	81.58	85.57	83.10	83.00	**85.77**	82.91	83.10	83.38
Stressed grass(C1)	84.1	83.65	98.21	85.06	84.02	87.34	**100**	82.61	99.53
Synthetic grass(C2)	**100**	**100**	98.42	99.60	99.80	**100**	**100**	**100**	**100**
Trees(C3)	93.09	93.09	97.73	91.57	91.95	94.26	96.88	93.37	**99.53**
Soil(C4)	**100**	99.91	96.5	98.86	99.62	98.42	99.91	99.72	98.86
Water(C5)	99.3	95.10	97.2	**100**	97.20	99.91	**100**	**100**	**100**
Residential(C6)	92.82	82.65	85.82	**96.64**	95.15	94.59	95.15	92.26	93.94
Commercial(C7)	82.34	81.29	56.51	88.13	83.29	91.81	93.92	91.55	**94.49**
Road(C8)	84.7	88.29	71.2	85.93	88.10	89.35	**91.31**	86.02	86.69
Highway(C9)	65.44	89.00	57.12	74.42	69.69	72.43	92.95	66.31	**97.97**
Railway(C10)	88.24	83.78	80.55	84.54	80.74	96.57	94.69	95.92	**98.39**
Parking Lot 1(C11)	89.53	90.39	62.82	95.39	89.82	95.60	89.43	94.24	**95.68**
Parking Lot 2(C12)	92.28	82.46	63.86	87.37	88.77	94.37	83.16	**98.60**	92.63
Tennis Court(C13)	96.76	**100**	100	95.14	94.33	99.86	**100**	**100**	**100**
Running Track(C14)	99.79	98.10	98.1	**100**	**100**	99.99	**100**	**100**	96.83
OA%	87.98	88.52	83.33	89.55	87.98	92.06	94.19	90.06	**95.24**
AA%	90.11	89.95	83.21	91.05	89.70	93.35	94.69	92.25	**95.86**
Kappa×100	86.98	87.59	81.88	88.71	87.02	91.42	93.69	89.24	**94.84**

For fair comparisons, the training and testing samples are the same for all compared algorithms. Two-branch CNN extracts spatial and spectral features from HS data, and extracts LiDAR image features from a cascade network. MDL-Middle and MDL-Late are foundational concatenation-based CNN models using intermediate and late fusion, respectively.

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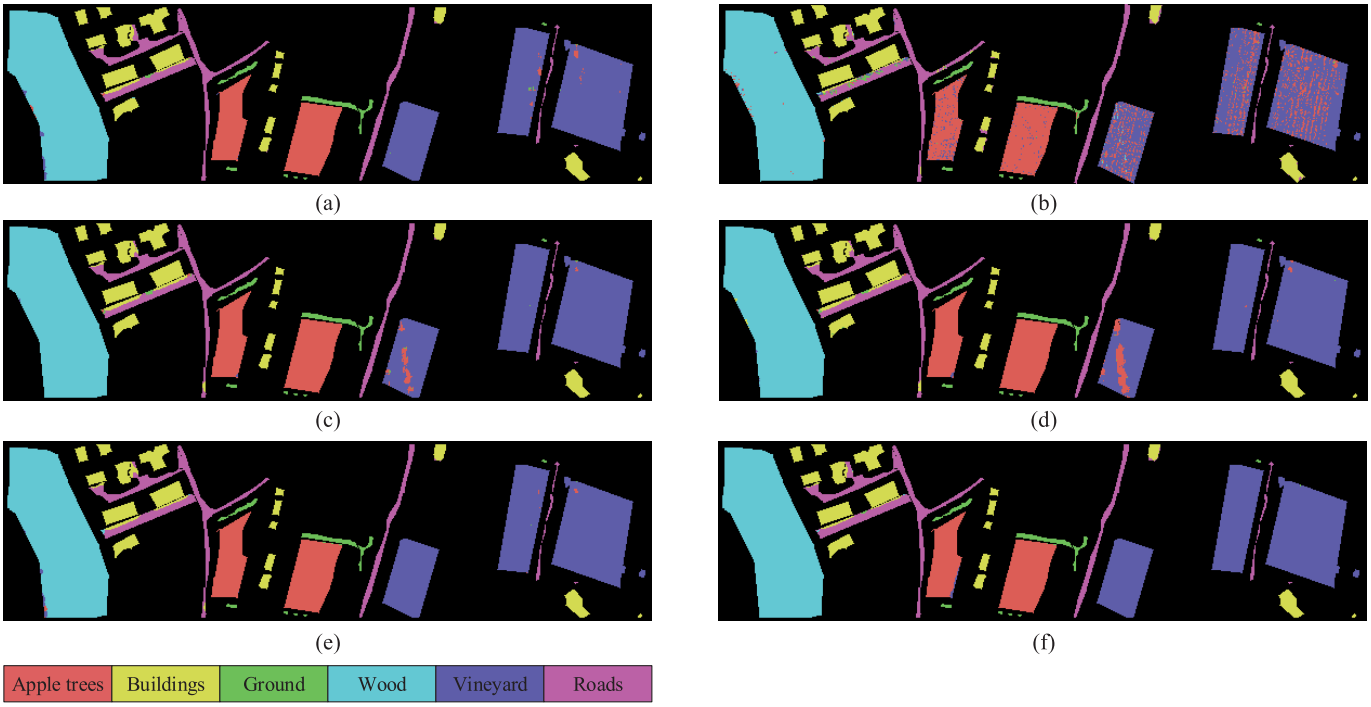


Fig. 8. Classification maps for Trento dataset. (a) Two-branch CNN. (b) EndNet. (c) MDL-Middle. (d) MDL-Late. (e) S^2ENet. (f) Proposed.

TABLE VI
QUANTITATIVE COMPARISON ON TRENTO DATASET OF DIFFERENT METHODS. BEST OUTCOMES ARE DENOTED IN BOLD

Category	Two-Branch CNN [22]	EndNet [44]	CNN-PPF [47]	MDL-Middle [25]	MDL-Late [25]	IP-CNN [45]	S^2ENet [46]	MAHiDFNet [24]	Proposed
Apple trees(C0)	98.07	88.19	90.11	**99.50**	99.73	99.00	99.65	98.90	98.86
Building(C1)	95.21	98.49	83.34	97.55	97.97	**99.40**	97.31	98.70	97.76
Ground(C2)	93.32	95.19	71.13	99.10	**100**	99.10	99.67	98.93	**100**
Woods(C3)	99.93	99.30	99.04	99.90	99.74	99.92	99.97	**100**	99.99
Vineyard(C4)	98.78	91.96	99.37	99.71	96.01	99.66	99.72	**100**	99.90
Roads(C5)	89.98	90.14	89.73	92.25	89.48	90.21	93.24	93.41	**94.64**
OA%	97.92	94.17	94.76	98.73	97.20	98.58	98.87	**99.03**	**99.03**
AA%	96.19	93.88	88.97	98.00	97.15	97.88	98.26	98.32	**98.52**
Kappa×100	96.81	92.22	93.04	98.32	96.28	98.17	98.50	98.70	**98.71**

EndNet achieves multimodal data fusion by forcing the fused features to reconstruct multimodal inputs, aiming to fuse more compactly. IP-CNN is designed with a bi-directional auto-encoder that can reconstruct HS and LiDAR data without the need for label information. CNN-PPF is initially proposed for the HSI classification task. S^2ENet proposed a cross-modal enhancement module based on an attention mechanism for spatial-spectral information interaction. MAHiDFNet developed a triple-branch CNN backbone to extract spectral, spatial and elevation features simultaneously.

Our proposed network consistently outperforms all advanced methods by a considerable margin in all pathways, whether it be OA, AA or kappa coefficient. Furthermore, it is evident that our method outperforms most other methods in terms of classification accuracy for the majority of classes, while for a few classes, other methods slightly outperform our method.

For Houston2013 dataset, in case of classes with similar shape characteristics, such as highway and railway, the proposed method yields substantial performance gains of 5.02% and 1.82%, respectively, compared to the second method. Our method also shows a notable increase for "tree" and "commercial" classes, the highest results of other methods in these two classes are 96.88% and 93.92%, respectively. Similarly, for Trento dataset, there is a significant improvement in the accuracy of the "ground" class and "roads" class.

Figs. 7 and 8 visually display the corresponding classification maps produced by these algorithms for Houston2013 and Trento datasets. We can visually see that the classification maps obtained from our proposed method have a tendency to have less noise and smooth transitions between categories. Fig. 7 shows that other methods have lower accuracy in classifying "tree" and "commercial" areas. Our approach also alleviates this problem to a large extent.

During the training process, we monitor the loss function to ensure convergence, and the loss-epoch curve is shown in Fig. 9. We also compare our algorithm with several state-of-the-art methods, including Two-branch CNN [22],

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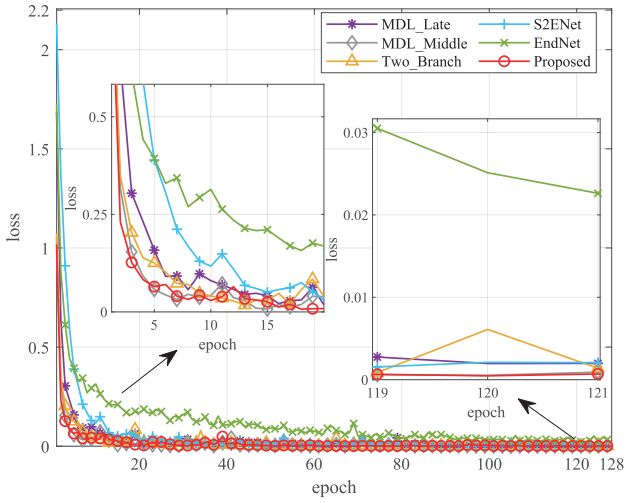


Fig. 9. Training loss curve with epoch on Houston2013 dataset. The proposed algorithm (red solid line) exhibits the fastest convergence and the lowest final loss, outperforming all other methods (dashed and dotted lines).

MDL-Late [25], MDL-Middle [25], S^{2}ENet [46] and EndNet [44]. We draw the loss-epoch curves of these models to compare their convergence performance. The results show that our proposed algorithm achieves the fastest convergence speed and the lowest final loss, as illustrated in Fig. 9. Adaptive adjustment of the backpropagation gradients after each batch, while maximizing the consistency between dominant modality features and integrated features, is the key that contributes to the superior performance of our algorithm. These results demonstrate the effectiveness of our proposed algorithm in solving pixel-by-pixel image classification tasks.

D. Ablation Study

To analyze the contribution of various components in the proposed model toward classification performance, we conduct experiments by removing certain parts of the model. The ablation study was conducted on Houston2013 and Trento datasets. Tables VII–IX show the experimental results, respectively, where ✓ denotes adding modules.

1) With/Without the DMGO Strategy: We comprehensively evaluate the effectiveness of our proposed gradient modulation strategy in our algorithm. The results in Table VII demonstrate that adding the DMGO strategy enhances the model's classification performance. Furthermore, to provide additional evidence of the superiority of the DMGO strategy, we conduct a comparative analysis against the gradient-blending (G-Blend) [35] method. This method adjusts the loss weight of different modalities based on their over-fitting phenomenon, which is quantified according to the different performances of the model on the training set and the validation set.

The experimental results of the two strategies are shown in Table VII. It can be seen that our gradient modulation strategy is superior to G-Blend on both public datasets. Specifically, G-Blend achieve an accuracy of about 94.60%, and DMGO further increased the accuracy by about 0.68%. The reason is that our DMGO strategy can directly adjust the back propagation gradient of each modality, which allows more efficient

TABLE VII
ABLATION STUDIES ON THE DMGO STRATEGY FOR DIFFERENT DATASETS. ✓ MEANS ADD MODULE

Dataset	DMGO	G-Blend [48]	OA	AA	Kappa
Houston2013	×	×	92.06	92.16	0.9139
	✓	×	**95.24**	**95.86**	**0.9484**
	×	✓	94.60	95.35	0.9414
Trento	×	×	98.49	97.71	0.9800
	✓	×	**99.03**	**98.52**	**0.9871**
	×	✓	98.64	97.90	0.9818

use of the information in each modality, resulting in better model performance. Compared with the step-wise G-Blend method, DMGO provides a more effective solution without training multiple models. This is particularly advantageous as training multiple models can be computationally expensive and time-consuming. In addition, DMGO is less dependent on the validation set compared to G-Blend, which means it is less affected by the representativeness of the validation set and can be more easily deployed in practical scenarios.

The effectiveness of the DMGO strategy can also be analyzed from the perspective of 2-D feature visualization. Specifically, we applied uniform manifold approximation and projection (UMAP) algorithm to embed features from MBE module into the 2-D space. The visualization results of the Houston2013 dataset extracted using DMGO and G-Blend are shown in Figs. 10 and 11, respectively. According to the results, the features of the same category become closer in the model with DMGO strategy. The reason may be that the DMGO strategy can directly adjust the backpropagation gradient of each modality, which is helpful to align the feature distribution of different modalities, so as to adjust the classification decision boundary accordingly. This alignment of feature distribution and decision boundary adjustment is not possible with the G-Blend method, as it only adjusts the loss weights of different modalities based on the overfitting-to-generalization ratio [35].

2) With/Without the MBE Module: The MBE module plays a crucial role in our proposed feature fusion framework. Experimental results in Table VIII highlight the efficacy of MBE module in enhancing feature extraction of HS and LiDAR data by generating interacted features. Specifically, the module uses self-attention masks from HS features to focus on their spectral-specific traits and cross-attention masks from LiDAR to enhance spatial representation. By extracting multimodal integrated features from multiple modalities, the MBE module significantly improves classification accuracy.

3) With/Without the FDC Loss Function: To show the importance of the suggested loss function, the FDC loss function was eliminated. The experimental results are shown in Table IX, and it is clear that the accuracy decreases when the FDC loss function is eliminated. The reason is that the FDC loss function maximizes the consistency between dominant modality features and integrated features, and enhances the model's ability to classify multimodal data.

4) Changing Loss Functions: In this work, we evaluated the effectiveness of different loss functions for RS image

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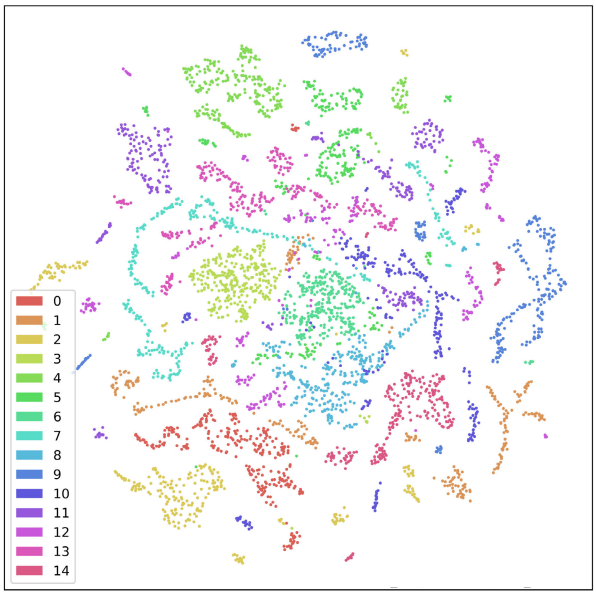


Fig. 10. Feature visualization with G-Blend method of Houston2013 dataset. (0)–(14) are labels which can be obtained from Table I.

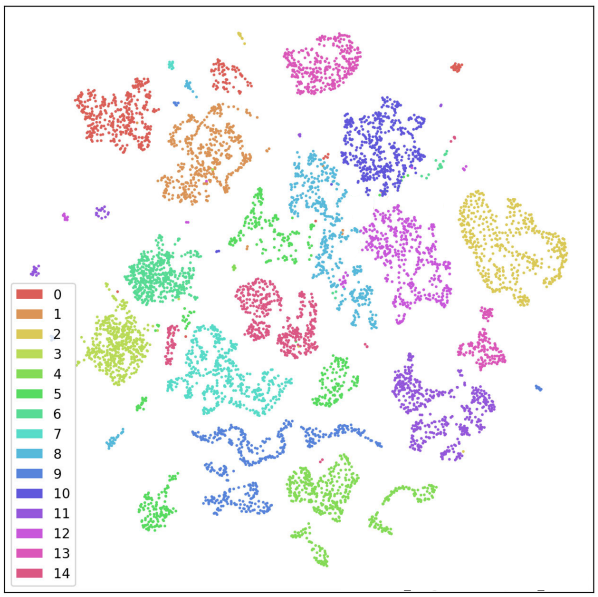


Fig. 11. Feature visualization with DMGO strategy of Houston2013 dataset. (0)–(14) are labels which can be obtained from Table I.

TABLE VIII
ABLATION STUDIES ON THE MBE MODULE FOR DIFFERENT DATASETS. ✓ MEANS ADD MODULE

Dataset	MBE Module	OA	AA	Kappa
Houston2013	×	94.07	95.00	0.9357
	✓	**95.24**	**95.86**	**0.9484**
Trento	×	98.53	97.71	0.9800
	✓	**99.03**	**98.52**	**0.9871**

classification task. Specifically, we replaced the proposed FDC loss with two widely-used loss functions in multiclass classification tasks, LovaszSoftmaxLoss [48] and FocalLoss [49]. Additionally, we compared the classification performance with

TABLE IX
ABLATION STUDIES ON THE PROPOSED LOSS FUNCTION FOR DIFFERENT DATASETS. ✓ MEANS ADD MODULE

Dataset	FDC Loss	OA	AA	Kappa
Houston2013	×	94.57	95.04	0.9411
	✓	**95.24**	**95.86**	**0.9484**
Trento	×	98.73	98.08	0.9800
	✓	**99.03**	**98.52**	**0.9871**

TABLE X
EFFECTIVENESS OF DIFFERENT LOSS FUNCTIONS FOR RS IMAGE CLASSIFICATION TASK ON HOUSTON2013 DATASET

Loss function	OA	AA	Kappa
LovaszSoftmaxLoss [48]	91.26	92.41	0.9051
FocalLoss [49]	94.21	94.81	0.9372
CE [46]	94.57	95.04	0.9411
Proposed FDC	**95.24**	**95.86**	**0.9484**

the traditional CE loss [46] function. Table X presents the experimental results, which indicate that our FDC loss surpasses other loss functions. Specifically, our FDC loss function shows improvements of 3.98%, 1.03%, and 0.67% compared to LovaszSoftmaxLoss, FocalLoss, and CE, respectively. In addition, the proposed loss function achieves higher AA and kappa coefficient compared to the other three loss functions on the Houston2013 dataset. The reason for such a large improvement is that our FDC loss constrains the distribution of the multimodal integrated feature to be similar to the dominant feature and makes the representation of each class as consistent as possible. Our FDC loss can boost the ability of our proposed method to capture more subtle variations between classes and achieve better classification results.

5) Changing the Proportion of Training Set: This section presents the results of an experiment to evaluate the classification accuracy of our proposed method on different percentages of the training set. We varied the training set size from 10% to 100% and evaluated the performance on Trento and Houston2013 datasets. The results of this experiment are summarized in Table XI. We observe that, for both datasets, the OA, AA, and Kappa coefficient generally increased as the proportion of the training set increases. This trend is expected since more labeled data can be used to train the classifier, leading to better performance. Specifically, for Trento dataset, our proposed method achieved an OA of 97.95% with 10% of the data used for training, which increased by 1.08% when the full dataset is used for training. Similarly, for Houston2013 dataset, the OA increased by 6.23% as the training set size increases from 10% to 100%. These results suggest that our proposed method benefits from using a larger proportion of the training set and can achieve high classification accuracy even with a small training set size. However, the performance improvement diminishes as the training set becomes larger, indicating that the proposed method may reach its optimal performance with a certain amount of labeled data.

6) Complexity and Operational Efficiency Comparison Among Various Models: To analyze the computational complexity of different models, we calculate the total parameters,

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TABLE XI
IMPACT OF TR SET SIZE ON CLASSIFICATION ACCURACY OF PROPOSED METHOD. TR SET STANDS FOR TRAINING SET

Dataset	Indicator	10% TR Set	20% TR Set	40% TR Set	70% TR Set	90% TR Set	100% TR Set
Trento	OA(%)	97.95	98.30	98.41	98.79	98.95	99.03
	AA(%)	97.73	97.70	97.54	98.12	98.38	98.52
	Kappa	0.9727	0.9774	0.9788	0.9839	0.9861	0.9871
Houston2013	OA(%)	89.01	92.57	93.35	93.85	94.38	95.24
	AA(%)	89.77	93.30	94.23	94.67	95.12	95.86
	Kappa	0.8807	0.9194	0.9277	0.9333	0.939	0.9484

TABLE XII
MODEL COMPLEXITY AND OPERATIONAL EFFICIENCY OF ALL METHODS ON HOUSTON2013 DATASET

Method name	ATT(s)	AIT(s)	Total Params	FLOPs
Two-Branch CNN [22]	275.2	73.5	74.928K	4.005M
MDL-Middle [25]	308.9	77.7	119.168K	5.339M
MDL-Late [25]	315.2	83.6	128.960K	5.484M
EndNet [44]	170.6	27.7	92.033K	184.640K
S^2ENet [46]	336.9	96.4	270.905K	32.775M
Our method	1272.5	206.1	388.584K	155.671M

FLOPs, average inference time (AIT), and average training time (ATT) of the proposed model and several advanced models, as shown in Table XII. It is evident that our model has a higher computational complexity compared to some other models listed in the table, resulting in longer ATT and AIT. However, we emphasize that the proposed model achieved noticeably better OA and other performance indicators, which justifies the increased computational costs. Moreover, we stress that the model was trained and evaluated on specific hardware configurations, and the calculation requirements may vary on other hardware. We provide the number of parameters and FLOPs of our model in the table to aid readers in comprehending the model's computational complexity of the model.

IV. CONCLUSION

This article proposes a dual-branch dynamic modulation network for HS and LiDAR data classification. Our model demonstrates superior fusion classification performance compared to several state-of-the-art multimodal classification methods, as evidenced by the experimental results on two publicly available HS-LiDAR datasets. It is verified that the DMGO strategy can dynamically and adaptively control the training process of each modality via gradient modulation. Moreover, it has been shown that the MBE module can extract integrated features of different modalities, thus promoting the improvement of classification accuracy. The FDC loss is designed to encourage the model to learn consistent information shared by different modalities, which can improve the consistency of features across modalities.

In the proposed model, the gradient modulation strategy optimizes the deep network for all channels of the dominant modality without discrimination. Our future work will extend the present study, i.e., selecting a few sensitive channels rather than all of them for optimization.

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